```
> restart; #The best fit for photons Planck's low

> with(PDEtools, dchange):with(plots):
> with(Units):
Automatically loading the Units[Simple] subpackage


> kernelopts(maxdigits);
```

$$38654705646 \tag{1}$$

```
> Digits:=10;
```

$$Digits := 10 \tag{2}$$

```
> params:={k= 1.380649*10^(-23)*Unit(J/K),rb=5.293*10^(-11)*Unit(m),he=
  4.135667697*10^(-15)*Unit(eV*s),
  ec= 1.602176634*10^(-19)*Unit(C),Tq=1.765*10^(-19)*Unit(s),
  c= 299792458*Unit(m/s),T=297*Unit(K),nu1=7.880979442*10^14*Unit(s^(-1)
  ),E1=3.259*Unit(eV)};
```

$$params := \left\{E1 = 5.221493608 \times 10^{-19}\ \text{J},\ T = 297\ \text{K},\ Tq = 1.765000000 \times 10^{-19}\ \text{s},\ c = 299792458\ \frac{\text{m}}{\text{s}},\right. \tag{3}$$

$$ec = 1.602176634 \times 10^{-19}\ \text{C},\ he = 6.626070096 \times 10^{-34}\ \frac{\text{m}^2\,\text{kg}}{\text{s}},\ k = 1.380649000 \times 10^{-23}\ \frac{\text{m}^2\,\text{kg}}{\text{s}^2\,\text{K}},$$

$$\left. nu1 = 7.880979442 \times 10^{14}\ \frac{1}{\text{s}},\ rb = 5.293000000 \times 10^{-11}\ \text{m}\right\}$$

```
> NV:=8*Pi*Tq*nu^2*he*nu/(exp(he*nu/(k*T) - 1));
```

$$NV := \frac{8\,\pi\,Tq\,\nu^3\,he}{e^{\frac{he\,\nu}{kT}} - 1} \tag{4}$$

```
> E:=diff(NV,nu);
```

$$E := \frac{24\,\pi\,Tq\,\nu^2\,he}{e^{\frac{he\,\nu}{kT}} - 1} - \frac{8\,\pi\,Tq\,\nu^3\,he^2}{\left(e^{\frac{he\,\nu}{kT}} - 1\right)kT} \tag{5}$$

```
> solve(E=0);
```

$$\{T = T,\ Tq = Tq,\ he = he,\ k = k,\ \nu = 0\},\ \{T = T,\ Tq = Tq,\ he = 0,\ k = k,\ \nu = \nu\},\ \{T = T,\ Tq = 0,\ he = he,\ k = k,\ \nu = \nu\},\ \left\{T = \frac{he\,\nu}{3\,k},\ Tq = Tq,\ he = he,\ k = k,\ \nu = \nu\right\} \tag{6}$$

```
> #Thus, for k=3/2*k*T we have the solution
> c:=299792458;nu1:=7.880979442*10^14;c/nu1;ec:= 1.602176634*10^(-19);
  k:= 1/2;E1:=3.259*ec;T:=297;Tq:=1.765*10^(-19);h := 6.62607015*10^
  (-34);he := h/ec;
```

$$c := 299792458$$

$$nu1 := 7.880979442 \times 10^{14}$$

$$3.804000000 \times 10^{-7}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$k := \frac{1}{2}$$

$$E1 := 5.221493650 \times 10^{-19}$$

$$T := 297$$

$$Tq := 1.765000000 \times 10^{-19}$$

$$h := 6.626070150 \times 10^{-34}$$

$$he := 4.135667697 \times 10^{-15} \quad (7)$$

```
> NV:=8*Pi*Tq*nu^2*he*nu/(exp(he*nu/(k*T) - 1));
```

$$NV := \frac{1.834552756 \times 10^{-32}\, \nu^3}{e^{2.784961412 \times 10^{-17} \nu - 1}} \quad (8)$$

```
> NV;
```

$$\frac{1.834552756 \times 10^{-32}\, \nu^3}{e^{2.784961412 \times 10^{-17} \nu - 1}} \quad (9)$$

```
> eval(eval(NV,{E(nu)=E__nu1,nu=nu1}),params);
```

$$2.387994260 \times 10^{13} \quad (10)$$

```
> E:=diff(NV,nu);
```

$$E := \frac{5.503658268 \times 10^{-32}\, \nu^2}{e^{2.784961412 \times 10^{-17} \nu - 1}} - \frac{5.109158634 \times 10^{-49}\, \nu^3}{e^{2.784961412 \times 10^{-17} \nu - 1}} \quad (11)$$

```
> solve(E=0);solve(5.503658268*10^(-32)*nu^2/exp(2.784961412*10^(-17)*nu
  - 1) - 5.109158634*10^(-49)*nu^3/exp(2.784961412*10^(-17)*nu - 1)=he*
  nu);
```

$$0., 1.077214207 \times 10^{17}$$

Warning, solutions may have been lost

$$0., 8.881647762 \times 10^{15} - 3.523994839 \times 10^{16}\, I \quad (12)$$

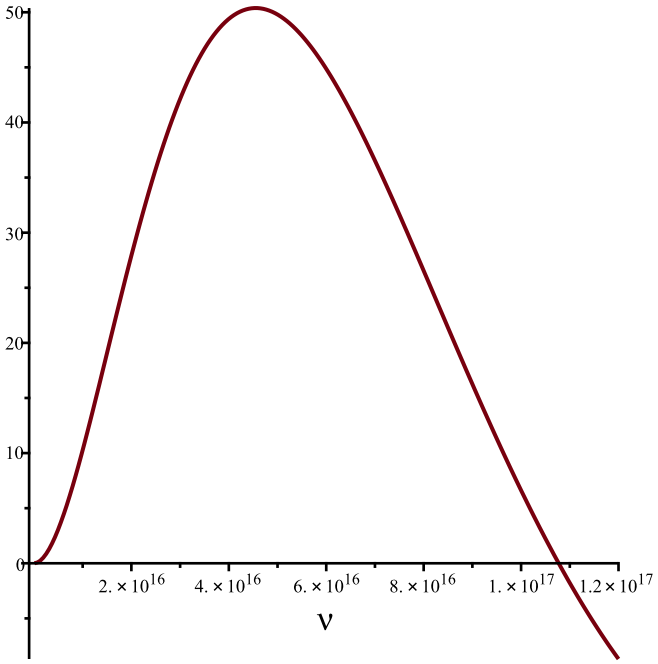
```
> series(-2.000000000*10^(-58)*(2.554579317*10^9*nu^2 - 2.751829134*
  10^26*nu)/exp(2.784961412*10^(-17)*nu - 1.),nu,9);
```

$$1.496049426 \times 10^{-31} \nu - 5.555253230 \times 10^{-48} \nu^2 + 9.669478676 \times 10^{-65} \nu^3 - 1.077164999 \times 10^{-81} \nu^4 + 8.749600280 \times 10^{-99} \nu^5 - 5.569668350 \times 10^{-116} \nu^6 + 2.908370852 \times 10^{-133} \nu^7 - 1.285666719 \times 10^{-150} \nu^8 + O(\nu^9) \quad (13)$$

```
> h1:=convert(%,polynom)/(1.6*10^(-18));
```

$$h1 := 9.350308912 \times 10^{-14} \nu - 3.472033269 \times 10^{-30} \nu^2 + 6.043424172 \times 10^{-47} \nu^3 - 6.732281244 \times 10^{-64} \nu^4 + 5.468500175 \times 10^{-81} \nu^5 - 3.481042719 \times 10^{-98} \nu^6 + 1.817731782 \times 10^{-115} \nu^7 - 8.035416994 \times 10^{-133} \nu^8 \quad (14)$$

```
> plot(E,nu=10^14..1.2*10^17);#The exact solution
```




```
> evalf(subs(nu=c/(3*10^(-9)),E));
```

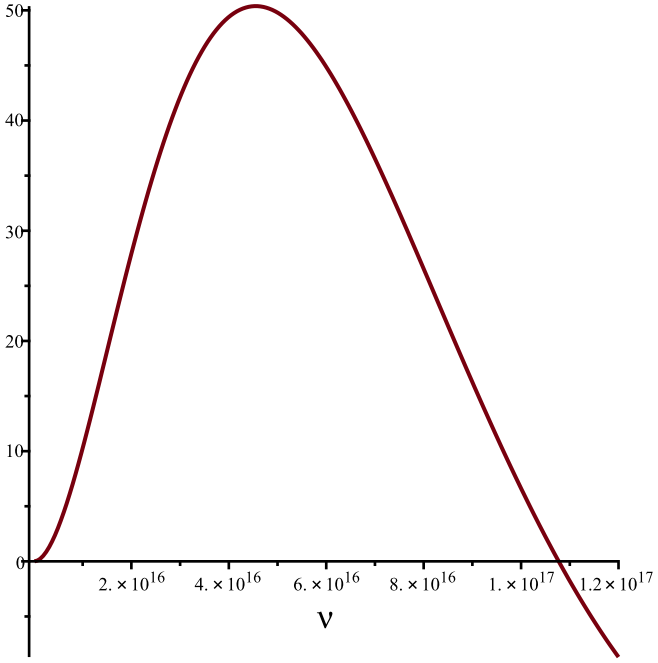
$$6.682779197 \tag{15}$$

```
> evalf(c/(3*10^(-9)));1/ec;
```

$$9.993081933 \times 10^{16}$$

$$6.241509074 \times 10^{18} \tag{16}$$

```
> plot(E,nu=10^14..1.2*10^17);
```




```
> #All we have to do is to fit the constant k=1/2 so for lambda=3.08nm
  and for lambda=380.4nm we have B=3.259 eV
```

```
> evalf(c/(9.993081933*10^16));evalf(c/(380*10^(-9)));evalf(subs(nu=c/
  (380.4*10^(-9)),E));evalf(subs(nu=c/(3.08*10^(-9)),E));
```

$$3.000000001 \times 10^{-9}$$

$$7.889275211 \times 10^{14}$$

$$0.09023714125$$

$$9.086107279 \tag{17}$$

```
> evalf(c/(3.08*10^(-9)));evalf(c/(9.733521364*10^16));
```

$$9.733521364 \times 10^{16}$$

$$3.079999999 \times 10^{-9} \quad (18)$$

```
> evalf(subs(nu=c/(345*10^(-9)),E));evalf(subs(nu=c/(315*10^(-9)),E));
  evalf(subs(nu=c/(10^4*10^(-9)),E));
```

$$0.1093758750$$

$$0.1307980197$$

$$0.0001343086142 \quad (19)$$

```
> restart;
> with(PDEtools, dchange):with(plots):
> with(Units):
Automatically loading the Units[Simple] subpackage

> kernelopts(maxdigits);
```

$$38654705646 \quad (20)$$

```
> Digits:=10;
```

$$Digits := 10 \quad (21)$$

```
> params:={k= 1.380649*10^(-23)*Unit(J/K),rb=5.293*10^(-11)*Unit(m),he=
  4.135667697*10^(-15)*Unit(eV*s),
  ec= 1.602176634*10^(-19)*Unit(C),Tq=1.765*10^(-19)*Unit(s),
  c= 299792458*Unit(m/s),T=297*Unit(K),nu1=7.880979442*10^14*Unit(s^(-1)
  ),E1=3.259*Unit(eV)};
```

$$params := \left\{E1 = 5.221493608 \times 10^{-19}\ \text{J},\ T = 297\ \text{K},\ Tq = 1.765000000 \times 10^{-19}\ \text{s},\ c = 299792458\ \frac{\text{m}}{\text{s}},\right. \quad (22)$$

$$ec = 1.602176634 \times 10^{-19}\ \text{C},\ he = 6.626070096 \times 10^{-34}\ \frac{\text{m}^2\,\text{kg}}{\text{s}},\ k = 1.380649000$$

$$\left. \times 10^{-23}\ \frac{\text{m}^2\,\text{kg}}{\text{s}^2\,\text{K}},\ \nu 1 = 7.880979442 \times 10^{14}\ \frac{1}{\text{s}},\ rb = 5.293000000 \times 10^{-11}\ \text{m}\right\}$$

```
> NV:=8*Pi*Tq*nu^2*h(nu)*nu/(exp(h(nu)*nu/(k*T) - 1));
```

$$NV := \frac{8\,\pi\,Tq\,\nu^3\,h(\nu)}{e^{\frac{h(\nu)\,\nu}{kT}} - 1} \quad (23)$$

```
> E:=nu*h(nu);
```

$$E := \nu\,h(\nu) \quad (24)$$

```
> ic:=h(7.889275211*10^14)=4.135667697*10^(-15);
```

$$ic := h\left(7.889275211 \times 10^{14}\right) = 4.135667697 \times 10^{-15} \quad (25)$$

```
> de:=diff(NV,nu)-h(nu)*nu;
```

$$de := \frac{24\,\pi\,Tq\,\nu^2\,h(\nu)}{e^{\frac{h(\nu)\,\nu}{kT}} - 1} + \frac{8\,\pi\,Tq\,\nu^3\left(\frac{\mathrm{d}}{\mathrm{d}\nu}\,h(\nu)\right)}{e^{\frac{h(\nu)\,\nu}{kT}} - 1} - \frac{8\,\pi\,Tq\,\nu^3\,h(\nu)\left(\frac{\left(\frac{\mathrm{d}}{\mathrm{d}\nu}\,h(\nu)\right)\nu}{kT} + \frac{h(\nu)}{kT}\right)}{e^{\frac{h(\nu)\,\nu}{kT}} - 1} \quad (26)$$

$$- \nu\,h(\nu)$$

```
> dsolve({de});
```

$$[\varnothing] \quad (27)$$

```
> #Thus, for k=3/2*k*T we have the solution
> c:=299792458;nu1:=7.880979442*10^14;c/nu1;ec:= 1.602176634*10^(-19);
  k:= 1/2;E1:=3.259*ec;T:=297;Tq:=1.765*10^(-19);he := h/ec;
```

$$c := 299792458$$

$$\nu 1 := 7.880979442 \times 10^{14}$$

$$3.804000000 \times 10^{-7}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$k := \frac{1}{2}$$

$$E1 := 5.221493650 \times 10^{-19}$$

$$T := 297$$

$$Tq := 1.765000000 \times 10^{-19}$$

$$he := 6.241509074 \times 10^{18}\, h \tag{28}$$

```
> NV:=8*Pi*Tq*nu^2*he(nu)*nu/(exp(he(nu)*nu/(k*T) - 1));
```

$$NV := \frac{27.68689003\, \nu^3\, h(\nu)}{e^{4.203036413 \times 10^{16}\, \nu\, h(\nu)} - 1} \tag{29}$$

```
> ic1:=h(9.489275211*10^16)=4.135667697*10^(-17);
```

$$ic1 := h(9.489275211 \times 10^{16}) = 4.135667697 \times 10^{-17} \tag{30}$$

```
> de:=diff(NV,nu)-h(nu)*nu;
```

$$de := \frac{83.06067009\, \nu^2\, h(\nu)}{e^{4.203036413 \times 10^{16}\, \nu\, h(\nu)} - 1} + \frac{27.68689003\, \nu^3 \left(\frac{d}{d\nu} h(\nu) \right)}{e^{4.203036413 \times 10^{16}\, \nu\, h(\nu)} - 1} \tag{31}$$

$$- \frac{27.68689003\, \nu^3\, h(\nu) \left(4.203036413 \times 10^{16}\, h(\nu) + 4.203036413 \times 10^{16} \left(\frac{d}{d\nu} h(\nu) \right) \nu \right)}{e^{4.203036413 \times 10^{16}\, \nu\, h(\nu)} - 1}$$

$$- \nu\, h(\nu)$$

```
> evalf(simplify(dsolve({de,ic1},h(nu),series)));
```

$$h(\nu) = 4.135667697 \times 10^{-17} - 4.060327648 \times 10^{71635144604154865} \left(\nu - 9.489275211 \times 10^{16}\right) \tag{32}$$

$$+ 3.287671071 \times 10^{143270289208309764} \left(\nu - 9.489275211 \times 10^{16}\right)^2 - 3.549395354$$

$$\times 10^{214905433812464663} \left(\nu - 9.489275211 \times 10^{16}\right)^3 + 4.310949299 \times 10^{286540578416619562} \left(\nu$$

$$- 9.489275211 \times 10^{16}\right)^4 - 5.584961425 \times 10^{358175723020774461} \left(\nu - 9.489275211 \times 10^{16}\right)^5 +$$

$$O\left(\left(\nu - 9.489275211 \times 10^{16}\right)^6\right)$$

```
> h0(nu):=convert(%,polynom);
```

$$h0(\nu) := h(\nu) = 3.852956650 \times 10^{71635144604154882} - 4.060327648 \times 10^{71635144604154865}\, \nu \tag{33}$$

$$+ 3.287671071 \times 10^{143270289208309764} \left(\nu - 9.489275211 \times 10^{16}\right)^2 - 3.549395354$$

$$\times 10^{214905433812464663} \left(\nu - 9.489275211 \times 10^{16}\right)^3 + 4.310949299 \times 10^{286540578416619562} \left(\nu$$

$$- 9.489275211 \times 10^{16}\right)^4 - 5.584961425 \times 10^{358175723020774461} \left(\nu - 9.489275211 \times 10^{16}\right)^5$$

```
> odetest(h0(nu), [de,ic1]);
Error, (in convert/rational) numeric exception: overflow
> odetest(h0(nu), [de,ic]);
```

Error, (in convert/rational) numeric exception: overflow

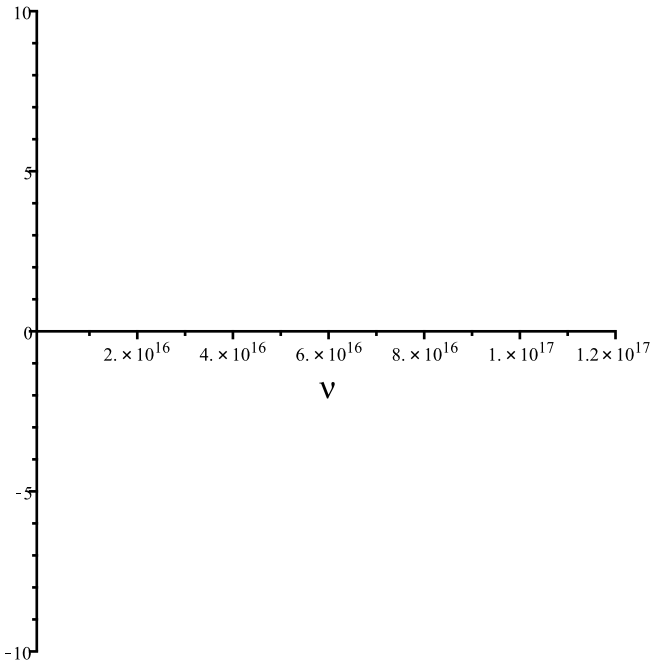
```
> evalf(c/(3.08*10^(-9)));1/ec;
```

$$9.733521364 \times 10^{16}$$

$$6.241509074 \times 10^{18} \quad \textbf{(34)}$$

```
> plot(h(nu),nu=10^14..1.2*10^17);
```

Warning, expecting only range variable nu in expression h(nu) to be plotted but found name h




```
> #All we have to do is to fit the constant k=1/2 so for lambda=3.08nm
  and for lambda=380.4nm we have E=3.259 eV
> h:=3.259/(9.733521364*10^16);
```

$$h := 3.348222989 \times 10^{-17} \quad \textbf{(35)}$$

```
> k=1.380649*297/1.6*10^(-5);
```

$$\frac{1}{2} = 0.002562829706 \quad \textbf{(36)}$$

```
> k=1.380649*10^(-23)*297/(1.60217*10^(-19));
>
```